

# ADRIAN MATIAS

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## PROFILE

Data Scientist / ML Engineer with 12 years shipping end-to-end machine learning from problem framing to production. Deep roots in core DS — ad-tech ranking & user acquisition (Appodeal, Chartboost), fraud detection, NLP (RavenPack), recommender systems, Bayesian Bandits / RL, and large-scale experimentation on Spark / Databricks. Current work in GenAI: LLM-as-judge product that scores software vendors, grounding raw vendor documentation into cited evidence, scoring engines. Owned the typed output contract and the correctness gate set. Approach driven by simplicity, first-principles thinking, and measurable business outcomes.

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## EXPERIENCE

### Senior AI Engineer

**Cavendish Professionals** → **PwC Contractor** cavendish-professionals | León, Spain – Remote | May 2026 – Present

Senior engineer on an LLM-as-judge product that scores software vendors. Created the human ground-truth lane alone at creation: 9 modules, 2,765 lines – domain-expert template parser, vendor matching, band placement, tier-weighted averaging and roll-up, which reconcile human evaluator scores against model scores.

#### KEY ACHIEVEMENTS

- Owned the typed output contract and the correctness gate set: type baselines per file and rule, a complexity ratchet, mutation tests and required status checks.
- Shipped 116 of 147 pull requests in ten weeks, median 3.0 hours to merge. Reviewed 52% of every pull request the rest of the team opened.
- Ran agentic development at production scale, measured it, and tuned turn count over model tier.

### Data Scientist → DS Technical Lead

**Appodeal**: appodeal.com | León, Spain – Remote | Dec 2024 – Jan 2026

Developed install prediction models for mobile user acquisition: cold-start, transformers, user embeddings, transfer learning, MLOps, production pipelines, parity resolution, experimentation, and monitoring. Progressed toward technical leadership based on ownership and strategic perspective.

#### KEY ACHIEVEMENTS

- Unified modeling across targets, advertisers, and platforms via attention/transformer architecture, increasing end KPI.
- Enabled seamless advertiser onboarding through novel cold-start techniques (custom PyTorch + transfer learning).
- Resolved parity mismatches and calibration gaps; productionized parity workflows with unit tests and decoupling in Databricks.
- Transfer-learning user embeddings achieving performance parity with orders-of-magnitude cost savings.

## Senior Data Scientist

**Quant AI Lab:** quant.global | León, Spain – Remote | Jul 2024 – Dec 2024

Designed and delivered end-to-end ML solutions for international clients; reviewed technical scope, aligned with stakeholders, and supervised model productivization.

### KEY CONTRIBUTIONS

- Led rapid turnaround of underperforming client ML system (utility fraud detection & ranking): diagnosed architectural issues, executed a surgical plan, restored accuracy/efficiency, and recovered client satisfaction.
- Discovered critical issues in data ingestion and feature-engineering pipelines.
- Led real-time credit card fraud classifier combining scalable ML with deep VAE. Open version: [github.com/adrianmatias/ml-sandbox/tree/master/tx\\_class](https://github.com/adrianmatias/ml-sandbox/tree/master/tx_class)

## Senior Data Scientist

**Chartboost:** chartboost.com | León, Spain – Remote | Jun 2023 – Jun 2024

Developed ad ranking and reinforcement learning systems for in-app game advertising; advanced DSP performance through large-scale experimentation and behavioral analysis.

### KEY ACHIEVEMENTS

- Evolved and monitored Bayesian Bandit RL system with policy evaluation tools; designed tailored online/offline evaluation frameworks.
- Built Two-Towers neural ranking algorithm based on recommender systems.
- Post-mortem analysis of adversarial DSP dynamics and model feedback loops.

## Machine Learning Engineer

**RavenPack** | León, Spain – Remote | Mar 2022 – Jan 2023

Researched, implemented, and maintained text analytics, NLU, and semantic search products on massive financial news data; productionized prototypes with emphasis on MLOps and reproducibility.

### KEY ACHIEVEMENTS

- Few-shot search personalization: simplified architecture, reduced error 30%.
- Active learning PoC for learn-to-rank: projected 300% annotation efficiency gain.
- Novel long-text fingerprinting algorithm for duplicate content detection.

### EARLIER ROLES

**Numbrs** · Data Scientist Aug 2020 – Oct 2021

Extended Deep Learning / Active Learning for transaction processing; scaled classifier across countries with GDPR-compliant pipelines.

**Stratio BD** · Data Scientist Aug 2019 – Aug 2020

Propensity/churn on Spark; BERT+graph email segmentation; multi-source weather DB (Elasticsearch/Flask).

**Digilant** · Machine Learning Engineer Jun 2015 – Apr 2017

Redesigned Consumer Persona ML model; Game Theory attribution prototype; co-directed Master's thesis.

**IEBS School** · Teacher Oct 2019 – Oct 2021

Webinars and masterclasses on real-time Big Data, AI, and ML applications.

**DOCOMO DIGITAL** · Data Scientist Apr 2017 – Aug 2019

Led fraud detection (99.8% specificity, <50 ms) and default-payment filtering; mentored juniors.

**2007–2015** · Tunnel Site Engineer & Maths & Sciences Tutor

Civil/tunnel engineering; university teaching: Math, Physics, Chemistry.

### FREELANCE

**Freelance Data Scientist / ML Engineer** | Mar 2022 – Jun 2025

On-demand projects in semantic search, time series forecasting, and portfolio optimization for financial analysis.

## SKILLS

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### Core Technologies

Python, FastAPI, Pydantic, PostgreSQL, PyTorch, Transformers, Azure OpenAI / Azure / AKS, AWS/S3, Databricks, PySpark, Pandas, SQL, Docker, MLflow, Git, GitHub Actions, AI-augmented tools (Ollama, llama.cpp, Hermes Agent)

### Engineering Practices

Clean code, typed contracts as source of truth, CI drift gates, OOP & FP, TDD / local e2e harnesses, reproducible pipelines, Agile / XP, production ownership

### Machine Learning & GenAI

LLMs, RAG, agentic systems, LangChain / LangGraph-style orchestration, document extraction & provenance, scoring engines, Deep Learning (attention, embeddings, transfer learning), Recommender Systems, NLP, RL / Bayesian Bandits, MLOps (parity, A/B, monitoring), Few-Shot / Active Learning, Ranking

### Consulting & Soft Skills

Multi-client consulting delivery, stakeholder communication, correctness gatekeeping, mentoring, business impact focus, simplification / unification

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## SELECTED COURSES & CERTIFICATIONS

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|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| - Claude Code: A Highly Agentic Coding Assistant: Coursera (2026)       | - Functional Programming in Scala Specialization: Coursera (9.2/10, 2017)              |
| - ML Engineering for Production (MLOps) Specialization: Coursera (2021) | - Bitcoin and Cryptocurrency Technologies: Coursera (9.5/10, 2017)                     |
| - Deep Learning with PyTorch: Siamese Networks (2022)                   | - Algorithms & Data Structures with Java (I, II): Coursera (8.5/10, 2014)              |
| - Deep Learning Specialization: Coursera (2016)                         | - Analytics Edge: Data Science & Predictive Models: MITx (9.3/10, 2014)                |
| - Structuring Machine Learning Projects: Coursera (2018)                | - Introduction to Programming & Data Science Using Python (I, II): MITx (8.7/10, 2014) |
| - Apache Spark for Big Data & Scalable ML: Coursera (9.9/10, 2015)      |                                                                                        |
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## SELECTED PROJECTS

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- **opo — AI-compiled exam preparation:** [opo.taoaieng.io](https://opo.taoaieng.io) Live paid product for Spanish public-sector exams: official law and past papers compiled once into a cited corpus of 28 topic pages and 660 tagged official items, then a 40-item diagnostic allocates the student's hours by official weight  $\times$  mastery deficit. Six proven CI gates block release; no claim ships without a citation.
  - **not the tao with adrian matias:** [youtube.com/@notTheTaoWithAdrianMatias](https://youtube.com/@notTheTaoWithAdrianMatias) Podcast exploring simplicity, consciousness, reality, machine learning, coding, sovereign AI.
  - **RAG Blog System:** [github.com/adrianmatias/ml-sandbox/tree/master/ragblog](https://github.com/adrianmatias/ml-sandbox/tree/master/ragblog) Retrieval-augmented generation pipeline integrating LLMs, vector embeddings, and web crawling.
  - **The Mirror:** <https://github.com/adrianmatias/the-mirror> Knowledge repository based on llm-wiki for personalized assistant and insight for agentic AI.
  - **Credit Card Transaction Fraud Classifier:** [github.com/adrianmatias/ml-sandbox/tree/master/tx\\_class](https://github.com/adrianmatias/ml-sandbox/tree/master/tx_class) Real-time fraud detection API (classical + unsupervised signals). Deploy-ready: FastAPI, Docker, RayServe.
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## EDUCATION

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- **Master's in Geological Engineering:** University of Oviedo (2010–2012)
  - **Degree in Mining & Energy Engineering:** University of León (2012)
  - **Bachelor in Mining Engineering:** University of León (2003–2007)
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*References available on request*